

Ammar Ahmed

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SUMMARY

Hands-on Mechanical Engineer with experience building and assembling UAV prototypes, integrating electronics, and delivering flight-ready demonstration models. Strong background in CAD, 3D printing (FDM), structural validation, and test-driven engineering. Familiar with ArduPilot and field-oriented iteration workflows.

PROFESSIONAL EXPERIENCE

Device Commissioning Engineer (Working Student) Nov 2024 – Present
WS Audiology GmbH, Germany

- Supported development and validation of hearing aid samples through structured subsystem and system-level testing.
- Created and maintained technical documentation, test procedures, and component-level records supporting series development.
- Built Power BI dashboards connected to SQL databases to visualize test results and evaluate performance metrics.

Research Officer (Mechanical Engineer) Apr 2023 – Jun 2024
National Aerospace Science & Technology Park (NASTP), Pakistan

- Manufactured and assembled UAV prototypes at scale for testing, supporting iterative development before next-stage builds.
- Built full demonstration drone models for pitching, including integration of electronics, propellers, actuators, landing gear, and structural components.
- Supported transition toward glass fiber reinforced composite structures, improving prototype robustness for the next development phase.
- Performed structural validation under aerodynamic and operational loading conditions to ensure component reliability.

Mechanical Design Engineer (UAV Prototyping) Oct 2022 – Apr 2023
Aero-Vision Technologies, Pakistan

- Designed foldable fixed-wing UAV structures and launch-tube deployment concepts for remote-area operations.
- Produced UAV prototypes using FDM 3D printing (PLA, ABS), including CAD pre-processing and printer operation/maintenance.
- Conducted CFD analysis to evaluate velocity/pressure profiles and estimate stall angle behavior for aerodynamic validation.
- Integrated mechanical assemblies with electronics and sensors; gained hands-on exposure to ArduPilot flight control workflows.

EDUCATION

M.Sc. Electromobility – ACES (E-train & Production Technology) 2024 – Present
Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU), Germany

B.Sc. Mechanical Engineering 2018 – 2022
National University of Sciences & Technology (NUST), Islamabad, Pakistan

PROJECTS

- **Regeneratively Cooled Bi-Propellant Rocket Engine (LOX/RP-1):** Thermal profiling and structural validation of high heat-flux components.
- **NUST Formula Student Development Team (FSDT):** Co-founded the team and led aerodynamics development using CFD-based design.
- **Solar Food Dryer (BSc Thesis):** Experimental test setup and validation; published in [Nature Food \(2023\)](#).

TECHNICAL SKILLS

- **CAD & Design:** SolidWorks (Advanced), DFMA, mechanical assemblies, tolerancing
- **3D Printing:** FDM prototyping (PLA, ABS), CAD pre-processing, printer operation, and maintenance
- **Analysis:** CFD (ANSYS Fluent), structural validation, durability and loading assessment
- **Testing & Data:** Test procedures, documentation, Power BI, SQL-based dashboards
- **Tools:** Python (Intermediate), ArduPilot (familiarity), Agentic AI
- **Languages:** English (Fluent), German (A2)